

Proteus: Adaptive Scale-Space Analysis for Learning Non-Linear Fuzzy Manifold Memberships (*Draft*)

Franz Wollang
Independent Researcher

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**This is a preliminary working version posted for discussion and feedback.
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Abstract

High-dimensional data often has structure that is both non-linear and scale-dependent: a region that behaves as one coherent mode at a coarse scale may split into several lower-scale modes, and the local intrinsic dimension can change across the support. Proteus is a two-stage framework for learning this structure as an explicit, queryable generative object. Stage 1 performs fast scale-space discovery with a distributional scaffold, identifying characteristic scales and recursively decomposing only where routed evidence supports further structure. Stage 2 lifts every accepted region of the resulting hierarchy into its own simplicial generative core, reconstructing probability fields on the learned complexes and attaching local non-linear warps only when geometric diagnostics and finite-data evidence agree. The output is a tree of complexes: each data point has a multiscale fuzzy membership trajectory through the tree, while likelihoods, samples, constraints, and custom region queries can be evaluated at any chosen resolution. Across the system, geometry proposes structural edits and a closed-form Dirichlet–multinomial evidence score accepts or rejects them, giving Proteus an evidence-grounded complexity control mechanism rather than a collection of independent heuristics.

1 Introduction

Motivation, contributions, and roadmap

Proteus begins from a simple premise: multiscale manifold structure should be learned as an explicit probabilistic object, not assembled after the fact from separate embeddings, clustering passes, density estimators, and geometric diagnostics. A useful representation should say which regions exist at which scales, what geometry supports those regions, and how likelihoods, samples, memberships, and constraints are evaluated on the same learned object.

The method therefore treats geometric structure as something to be proposed locally and accepted statistically. Stage 1 builds a fast distributional scaffold that accumulates local moment and transition evidence while searching for characteristic scales. Stage 2 lifts the supported regions into simplicial generative cores, reconstructs probability fields on those complexes, and adds local non-linear capacity only when both geometry and finite-data evidence warrant it. The introduction motivates this design, states the output object, and summarizes the paper’s main claims before Section 2 describes the system end to end.

1.1 Motivation and Gap

Many data sets are easier to understand as geometric objects than as flat vectors: samples concentrate near curved supports, local dimension can vary, and meaningful modes may appear or disappear depending on the scale at which the data are viewed. A useful model in this setting should do more than embed points. It should expose the learned geometry, identify the scales at which structure is statistically supported, and provide a generative interface for likelihoods, samples, and constrained queries.

Existing approaches each capture part of this picture. Embedding and manifold-learning methods can reveal geometry but usually do not produce an explicit probability model. Deep generative models can fit complex densities but often hide the geometry behind a parameterization that is hard to inspect or edit. Growing graph and mesh methods build explicit scaffolds, but their growth and pruning rules are often heuristic, making it difficult to distinguish true structure from finite-sample noise. Proteus targets the intersection of these goals: an explicit multiscale geometry with a probabilistic interface and an evidence-based rule for deciding when structure should be added, removed, or upgraded.

1.2 What Proteus Delivers

Proteus learns a hierarchy of locally scaled geometric models. Stage 1 builds a sparse scaffold, accumulates local evidence, and searches over variance caps to find characteristic scales. When a scale reveals supported substructure, Affinity Propagation on the scaffold graph defines child regions and the controller recurses. Stage 2 then lifts each region of this recursion tree into a simplicial complex at its own selected scale, equips it with a probability field, and optionally adds local non-linear warps where a piecewise-linear approximation is insufficient.

The final object is a *tree of complexes*. A query point descends through this tree, receiving a scalar fuzzy membership score at each region on its path. These scores form a multiscale membership trajectory: a compact, downstream-facing representation of where the point sits in the hierarchy and how close it is to the mode of each region. The same learned object also supports density evaluation, sampling, conditional synthesis, and region-level semantic queries.

1.3 Contributions

The main contribution is an architecture that separates fast multiscale discovery from evidence-grounded refinement into explicit simplicial generative cores. Stage 1 supplies the scale controller: it selects characteristic scales through a cluster-local response $\Phi_C(\tau)$ and recurses only when routed evidence supports additional structure. Stage 2 preserves the resulting hierarchy as a tree of complexes, one per accepted recursion-level region, so coarse modes and fine submodes remain queryable rather than being overwritten by later refinement.

On top of this representation, Proteus defines a closed-form multiscale fuzzy membership trajectory: each per-region score is a Gaussian relative-density in that region’s locally Gaussianized frame. The same model also introduces a single evidence gate for structural edits, implemented as a closed-form Dirichlet–multinomial marginal likelihood under a dimension-adaptive BDeu prior, and a dual-flow reconstruction that turns accepted, settled complexes into probability fields for likelihood evaluation, semantic queries, and conditional synthesis.

1.4 Roadmap

Section 2 gives the system-level overview and names the learned artifacts. Section 3 describes Stage 1 scaffold construction and scale discovery. Section 4 explains Stage 2 lifting, simplex equilibrium, torsion diagnostics, safe edits, and optional warps. Section 5 presents the evidence gate used for structure selection. Section 6 describes fuzzy memberships, densities, sampling, and

queries. Section 7 summarizes the theoretical and operational status, followed by evaluation, limitations, and conclusion. Detailed derivations, defaults, proofs, algorithms, and implementation protocols are collected in the Supplementary Information (SI).

1.5 Related Work

Proteus sits between manifold learning, explicit geometric modeling, and probabilistic generative inference. It shares the goal of local structure discovery with self-organizing maps, Growing Neural Gas [5], and related adaptive graph methods, but replaces most heuristic growth decisions with a common evidence criterion. It shares with computational topology [4] and discrete-geometric methods [1] the desire for an inspectable geometric object, but adds a learned probability field and query interface. Its scale-selection logic follows the scale-space tradition [7, 8], while its implementation relies on approximate-neighbor search [9]. Finally, Proteus treats flexible invertible maps such as Glow, RealNVP, and neural spline flows [6, 2, 3] as localized upgrades to an explicit geometric core rather than as the sole model class.

2 Proteus Overview

The learned object and the system contract

This section names the objects that recur throughout the paper and fixes the high-level contract between them. The purpose is not to give the full mathematical specification, which lives in the SI, but to make the architecture readable before the individual stages are introduced. In particular, we separate the learned artifacts from the empirical signals that train them, and we distinguish the online evidence used for structure selection from the reconstructed fields used for inference.

2.1 At a Glance

Proteus is organized around a repeated control pattern: *propose*, *test*, and *settle*. Fast geometric diagnostics propose local changes, a finite-data evidence score decides whether those changes are statistically warranted, and continuous state is allowed to re-equilibrate before the next batch of proposals. This pattern appears at multiple levels: Stage 1 proposes scaffold splits and prunes; the scale controller proposes recursive decomposition; Stage 2 proposes simplicial edits and local warps; and the same evidence gate arbitrates all accepted topology changes.

The two stages have different responsibilities. Stage 1 is the cheap discovery stage: it builds graphs, tracks local moments, and finds scale-dependent cluster regions. Stage 2 is the high-fidelity refinement stage: it turns each accepted region into a simplicial complex with a probability field and an inference interface. The separation keeps scale search inexpensive while reserving detailed density reconstruction for regions and scales that survived the evidence tests.

2.2 Learned Artifacts

Proteus produces several learned artifacts that together define the model rather than separate post-processing products.

Multiscale scaffold hierarchy. The first artifact is a recursion tree of node–edge graphs, each stabilized at a locally selected characteristic scale and carrying local moment summaries plus transition evidence.

Tree of simplicial complexes. For each accepted region of the scaffold hierarchy, Proteus constructs a simplicial complex $\mathcal{K}_C$ that represents that region at its own scale τ_C^* .

Per-region probability fields. Each complex carries simplex masses and reconstructed face-pressure fields that support density evaluation and sampling within that region.

Multiscale fuzzy membership trajectory. A query point receives a path of per-region membership scores $\mu(\mathbf{x})$ from root to leaf, exposing both hierarchical identity and continuous closeness to each region’s mode.

Optional localized warps. Invertible maps are attached only where torsion diagnostics and evidence tests show that a piecewise-linear representation needs additional non-linear capacity.

2.3 Three Fields and Where They Live

Three numerical fields are easy to conflate, so we name them up front. All are owned per recursion-tree region C .

- **Transition counts** $\{n_{i \rightarrow j}^{(C)}\}$. These are empirical evidence accumulated by routed neighbor assignments. They are written during sample processing and read by the structure-selection gate.
- **Simplex masses** $\{m_s^{(C)}\}$. These are latent compositional variables on the simplices of $\mathcal{K}_C$. They describe how much probability mass each simplex carries once a topology is accepted and settled.
- **Face pressures** $\{p_f^{(C)}\}$. These are reconstructed interpolation fields on simplex facets. Empirical face tallies are accumulated online, but the conservative pressure field is solved after topology selection.

The counts are the direct finite-data signal for accepting or rejecting structure. The masses and pressures are derived objects used by the generative and inference interface; they are not independent evidence terms for accepting the same topology.

2.4 The Pipeline Contract: Propose, Test, Settle

The pipeline contract is deliberately simple.

1. **Propose.** Geometry and statistics identify local strain: excessive variance, weak links, over-complete stars, high torsion, or insufficient warp capacity.
2. **Test.** Candidate structural changes are scored against the current topology using the local evidence gate. Weakly supported changes are rejected even if a geometric diagnostic fired.
3. **Settle.** After accepted edits, continuous state is re-equilibrated so the next decision is made against stable statistics rather than transients.

This contract is the main guard against uncontrolled growth. Geometry can suggest where the model is strained, but finite-data evidence decides whether more structure is justified.

2.5 What Is Deferred to the SI

The main paper focuses on architecture, intuition, and the minimum algorithmic detail needed to understand the model. The SI owns the mathematical and operational specification: derivations, defaults, gate cadences, algorithm boxes, proof sketches and theorem statements, numerical safeguards, logging requirements, and implementation protocols. When the same object appears in both documents, the paper gives the reader-facing interpretation and the SI gives the canonical definition.

Architecture overview placeholder: Stage 1 scaffold hierarchy, Stage 2 tree of complexes, evidence gate, dual-flow reconstruction, and inference interface.

Figure 1: Proteus architecture overview. The final figure should show the separation between scale discovery, evidence-gated refinement, probability reconstruction, and multiscale inference.

3 Stage 1: Scaffold and Scale Discovery

Fast distributional discovery before high-fidelity refinement

Stage 1 is the discovery layer of Proteus. Its job is to build a cheap, evidence-bearing scaffold before the method commits to the more expensive simplicial refinement of Stage 2. The same mechanism serves two purposes: at a fixed scale it learns local representatives, moments, and transition counts; across scales it identifies characteristic resolutions and decides where recursive decomposition is justified.

This section first describes the fixed-scale scaffold and the node statistics it maintains, then explains how the scale controller turns a family of stabilized scaffolds into a hierarchy of regions. The output is not yet the final generative model. It is the set of scale-selected regions, graphs, counts, and summaries that Stage 2 will lift into queryable complexes.

3.1 Operating Domain and Cluster Definition

Proteus is defined on a noise-positive operating domain rather than on an idealized zero-noise manifold limit. The Stage 1 theory assumes three axioms:

$$\sigma_{\text{noise}} > 0,$$

$\rho(\mathbf{x})$ exists as a probability density on the observation space,

$$\rho(\mathbf{x}) = u(\mathbf{x}) + \sum_{i=1}^m f_i(\mathbf{x}), \quad u(\mathbf{x}) \geq u_{\min} > 0.$$

The first states that every observational setting carries nonzero stochastic spread; the second fixes a continuous latent object on which scale-space reasoning is well defined; the third states that the density decomposes into a strictly positive background floor plus localized excess features. The detailed operating-domain consequences are derived in [SI S0.6](#).

The background floor is not operationally a single global nuisance constant. At the root, the relevant contrast reference is the ambient floor, but once recursion isolates a region, detectability and resolvability are governed by that region’s *local* floor. Stage 1 therefore treats noise as a tracked field on the scaffold: local residual statistics and settled catchments provide the quantities against which feature contrast is judged.

Definition 3.1 (Cluster at scale v). On the scaffold graph at variance scale v , a cluster is a non-empty node set S satisfying:

1. **Coherence:** its mean internal evidence exceeds its mean boundary evidence, equivalently $Q(S; v) > 0$;
2. **Extent:** its empirical variance is not sub-resolution at that scale, $\hat{v}_S \geq v$;
3. **Maximality:** no adjacent candidate region couples to S as strongly as either region couples internally.

This definition is intentionally broad enough to cover compact bump-like modes, extended manifold clusters such as noisy tubes or sheets, and hierarchical coarse-scale composites. Pure background tissue is the non-cluster case: it never develops positive internal contrast over its own boundary. The derivation of the definition from the operating-domain axioms is given in [SI S2.6](#).

3.2 Fixed-Scale Scaffold Construction

At a fixed variance cap τ , Stage 1 builds a sparse node–edge scaffold of the data distribution. Nodes represent local regions of support, and Hebbian edges represent repeated local co-activation under nearest-neighbor routing. Unlike a pure geometric graph, each node also stores distributional state: hit mass, residual moments, variance diagnostics, and local principal directions. The scaffold is therefore both a geometric object and a finite-data evidence accumulator.

The fixed-scale run is intentionally cheap. Each sample updates only a small rank-ordered neighborhood of nodes, with geometrically decaying weights by neighbor rank. This gives nearby nodes partial credit, stabilizes hit counts, and avoids making the BMU a brittle winner-take-all statistic. Directed transition counts $n_{i \rightarrow j}$ are first/second-nearest-neighbor co-activation tallies: they record how often local routing from node i points toward neighbor j . These counters later become the evidence consumed by clustering and structure selection.

Edges follow the usual GNG/Hebbian convention at a high level. When two nearby nodes are repeatedly co-activated by incoming samples, the scaffold creates or refreshes the edge between them if it is absent, and increments its directed transition counters. Edges whose support remains weak after a fair observation window become prune candidates, but deletion is delayed until the statistical gauntlets and evidence gate agree that the edge is not carrying meaningful local adjacency.

3.3 Node State and Variance Cap

For node i , Stage 1 tracks an exponentially weighted residual mean $\mathbf{m}_i$, second moment $\mathbf{s}_i$, derived local variance σ_i^2 , hit mass h_i , and an Oja direction for dominant unresolved variance. The variance cap $\tau_{\text{local},i}$ is the operational scale constraint: if a node’s local variance persistently exceeds its cap, the scaffold is under-resolved near that node and a split proposal is queued. Conversely, nodes or links with weak evidence and low variance become prune candidates.

Node motion is split into fast distributional tracking and slower geometric adjustment. Moment estimates update continuously at the EWMA rate $\alpha = \ln 2/k$, but node positions move only through deferred nudges, and the rates governing that motion are tied directly to the variance cap.

The adaptation speed at each node is set by how far its local variance is from the cap. Specifically, the variance-correction rate

$$\eta_{\text{GNG},i} = \frac{\ln 2}{2k} \left(1 - \frac{\sigma_i^2}{\tau} \right)$$

is positive when the node is under-cap (the node can still absorb displacement) and zero at the cap boundary (the node is at its resolution limit and should propose a split rather than move). This makes adaptation self-regulating: nodes in well-resolved regions are nearly stationary, while nodes explaining too much heterogeneity adjust quickly.

Position updates are further gated by a displacement threshold. The nudge accumulator is driven by the EWMA residual mean $\mathbf{m}_i$ rather than by each raw sample residual, so random per-sample directions are filtered before they can change geometry. Each node accumulates a deferred nudge $\mathbf{a}_i$, and the position $\mathbf{w}_i$ is updated only when $\|\mathbf{a}_i\|$ exceeds

$$\delta_{\min} = \kappa(1 - r)\sqrt{\tau}, \quad \kappa = 0.5 \text{ (default)},$$

where $r = 1/\sqrt{2}$ is the scale-grid ratio. This prevents high-frequency jitter from destabilizing the ANN index and ensures that realized moves are resolvable at the current scale. The centering rate that produces roughly one realized move per k updates is

$$\eta_{\text{cent}} = \frac{\kappa(1 - r)}{k},$$

coupling position dynamics to the same neighborhood window and grid spacing used by the EWMA and the scale controller. The full derivation under local isotropy assumptions is in [SI S2.3](#).

When σ_i^2 exceeds $\tau_{\text{local},i}$, the excess—termed the *variance surplus*—is the per-sample KL deficit between the local catchment density and the best cap-admissible Gaussian; integrated over routed hits, it is the evidence that the current resolution is insufficient and a split along $\mathbf{u}_i$ is warranted. A split refines the underlying mixture by partitioning one component into two; because the per-sample partition membership was not stored, the parent’s moment estimates do not transfer to the children identifiably. To recover identifiability streaming-side, Proteus optionally maintains a pair of *partition-aligned shadow EWMA*s along $\mathbf{u}_i$, so each child inherits real sufficient statistics rather than zeros at split time, while remaining $O(d)$ per node. The variational reading of the variance cap, the refinement-admissibility principle that classifies per-node state under refinement, and the shadow-moment construction are derived in [SI S2.3.1](#) and [S2.3.2](#); the matching transition-count rule appears in [SI S4.3–S4.4](#).

Stage 1 also maintains a per-node intrinsic-dimension estimate $d_{\text{final},i}$, derived from the local graph degree and smoothed over nearby scaffold nodes. Within a single scaffold run, the variance cap is uniform: $\tau_{\text{local},i} = \tau$ for every node. Dimensional heterogeneity within a region is resolved not by per-node cap modulation but by the scale response (S2.5) peaking at different τ for clusters of different intrinsic dimension, and by recursion giving each cluster its own scale search in its own PCA frame. The per-node estimate $d_{\text{final},i}$ enters the scale-response normalization, junction detection, and the dimension of simplices later built from the scaffold, but does not modulate the split-decision cap.

3.4 Stabilization at a Fixed Scale

A fixed- τ scaffold is considered usable only after its local statistics stabilize. Operationally, the controller monitors hit-mass dispersion, variance-cap violations, deferred-motion activity, and queued proposal rates over an equilibration window. Grid points that fail to stabilize are excluded from scale peak detection or given an expanded budget before being declared unreliable. This avoids treating transient growth behavior as a scale-space signal.

3.5 Scale Response and Characteristic Scales

For each current region C , the controller evaluates a cluster-level response $\Phi_C(\tau)$ over a geometric grid of candidate variance caps. Intuitively, $\Phi_C(\tau)$ is large when the scaffold at scale τ resolves the region’s support without over-refining it into redundant interior nodes. It is formed from per-node Lindeberg-normalized responses $R_i(\tau)$ summed over the cluster; those per-node responses are used for scale selection, not for within-cluster density evaluation.

Candidate characteristic scales are found by bracketing local maxima of $\Phi_C(\tau)$ using centered second differences, then refining promising brackets with a lightweight optimizer on $\log \tau$. A companion support trace checks whether the support volume is undergoing the expected plateau transition near the response peak. The selected scale τ_C^* is the scale at which the region is handed either to recursion or to Stage 2 refinement.

The discovery objective is explicitly coarse-to-fine: Stage 1 should identify features in decreasing characteristic scale, as close to each feature’s best operating scale as possible without overshooting into an already-dying regime. This means (i) coarse composite structures are established before their children are sought, (ii) the selected operating scale should lie inside the feature’s admissible persistence interval, (iii) child birth and parent death are dual recursion signals, and (iv) a feature should not be forced to survive below its useful operating scale merely because finer structure exists downstream. The characteristic-scale relations and parent/child handoff are formalized in [SI S2.6.2](#).

Scale-response placeholder: $\Phi_C(\tau)$ peak, support-trace plateau, and selected τ_C^* .

Figure 2: Characteristic-scale selection. The final figure should show how the cluster-local scale response and support trace identify the operating scale for a region.

Proposition 3.2 (Detectability above tissue). *For an isotropic Gaussian mode of intrinsic width σ_i , excess height H_i , and local operating floor u , detectability at smoothing scale τ requires*

$$\frac{H_i}{u} \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau} \right)^{d/2} \geq \kappa_{\text{det}}.$$

Equivalently, the maximum admissible local floor is

$$u \leq \frac{H_i}{\kappa_{\text{det}}} \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau} \right)^{d/2}.$$

Proposition 3.3 (Clean resolvability of two modes). *Two equal-width isotropic Gaussian modes of common width σ , separated by Δ , remain bimodal after smoothing by variance τ only if*

$$\Delta > 2\sqrt{\sigma^2 + \tau}.$$

At the natural operating scale $\tau = \sigma^2$, this gives the characteristic-scale resolution threshold

$$\Delta > 2\sqrt{2}\sigma.$$

These two inequalities anchor the scale controller. Proposition 3.2 states how much local tissue a feature can tolerate before losing contrast; Proposition 3.3 states how far neighboring features must be separated before a legitimate split is even possible. Their derivations and the bridge to the graph-local Q criterion are given in [SI S2.8](#).

3.6 Cluster Identification and Recursive Decomposition

After a characteristic scale is selected, the stabilized scaffold is clustered by Affinity Propagation on its Hebbian graph. Similarities are computed from smoothed transition statistics, so clusters reflect persistent routed adjacency rather than raw Euclidean proximity alone. Each AP cluster becomes a candidate child region; samples are routed to the child owning their BMU at the selected scale. Before that child begins its own scale search, the inherited intrinsic-dimension estimates define a local PCA frame for the child data and warm-start nodes (SI S1.4). Marginal Gaussianization and any later region or patch warps are fitted in this child PCA frame, making those transforms cheaper without dropping information retained by the child projection audit.

Recursion is evidence-limited. A child is expanded only if it has enough routed evidence, a meaningful scale-response peak in the searched band, and remaining recursion budget. Otherwise it is declared terminal and passed to Stage 2 at its current scale. This is the main mechanism preventing combinatorial explosion: without both samples and a distinguished scale, the hierarchy stops growing.

3.7 Stage 1 Algorithm

Algorithm 1 summarizes the scale-discovery loop at the level needed for the main text; Figure 2 illustrates the response trace used to select a characteristic scale.

Algorithm 1 Stage 1 scale discovery (paper view)

```
1: procedure DISCOVERREGION( $X_C$ )
2:   for candidate scales  $\tau$  on a geometric grid do
3:     Run the Stage 1 scaffold to stabilization at fixed  $\tau$ 
4:     Record  $\Phi_C(\tau)$  and support trace  $V_C(\tau)$ 
5:   end for
6:   Select characteristic scale  $\tau_C^*$  from robust response peaks
7:   Cluster the stabilized Hebbian graph by AP on smoothed transition evidence
8:   for each child cluster do
9:     if sample floor, response peak, and recursion budget pass then
10:      Project child data and nodes into the intrinsic-dimension PCA frame
11:      Recurse on the child region
12:     else
13:      Mark the child terminal for Stage 2 lifting
14:     end if
15:   end for
16: end procedure
```

3.8 Outputs Handed to Stage 2

For every accepted region C , Stage 1 hands Stage 2 a stabilized scaffold graph, the selected scale τ_C^* , node positions and local moment summaries in the region’s active working frame, intrinsic-dimension estimates, routed hit counts, directed transition counts, AP cluster ownership, and the region’s position in the recursion tree. The active working dimension D_{subspace} may differ by level because parent-to-child recursion applies the intrinsic-dimension PCA transfer before the next scale search. Stage 2 treats this as the warm start for a higher-fidelity generative core rather than rebuilding geometry from scratch.

4 Stage 2: Simplicial Generative Cores

From scaffold regions to queryable complexes

Stage 2 takes the scale-selected scaffold hierarchy from Stage 1 and turns it into the structure on which both refinement diagnostics and inference operate. The scaffold is good at routing evidence and proposing coarse structure, but it has no volume-bearing cells, no simplex-level mass field, and no natural domain for conservative density reconstruction. By lifting each accepted region into a simplicial complex at its own selected scale, Stage 2 creates the local cells needed to estimate simplex-volume hit mass, refine intrinsic-dimension estimates, accumulate face-pressure evidence, and measure torsion-like residual structure.

Those diagnostics are also what make the optional non-linear maps trainable in a controlled way: face pressures and torsion identify where the piecewise-linear complex is strained, while simplex-local statistics provide the geometry on which a local warp can be fitted and judged. The important design choice is that refinement does not collapse the hierarchy to its leaves. Coarse regions, intermediate modes, and fine submodes all remain explicit complexes, each with its own geometry, diagnostics, and probability field. This section explains why every accepted region is lifted, how the graph becomes a complex, and how simplex equilibrium and torsion diagnostics govern refinement before the evidence gate decides which structural changes survive.

4.1 Scope: A Tree of Complexes

Stage 2 is applied to every accepted region in the Stage 1 recursion tree, not only to the leaves. For each region C , the stabilized scaffold at scale τ_C^* is lifted to a complex $\mathcal{K}_C$ and equipped

with its own mass field and probability reconstruction. This matters because a coarse region may be a meaningful mode even if it later decomposes into finer children. The parent and child complexes represent different scale-smoothed views of the same data, so both should remain available for queries.

The result is a tree of complexes. A root complex captures the broad distribution, intermediate complexes capture coarse modes, and leaf complexes capture the finest accepted resolution. Inference operations choose a level explicitly or traverse multiple levels through the fuzzy membership trajectory.

Sibling stitching within a level. Vertices on the boundary between adjacent sibling regions at the same level are physically shared (the same Stage 1 nodes appear on both sides), and Hebbian links crossing a sibling boundary persist as bridge edges. Two variants govern whether cross-boundary cliques are materialized as simplices:

- *Disjoint siblings with bridge edges (default):* each $\mathcal{K}_C$ is built only from its own interior cliques; cross-boundary edges remain in the topology and route evidence but do not produce simplices spanning two regions. Each complex is a manifold-with-boundary at its own scale, and inter-sibling flux shows up in the dimensionality-junction signature (SI S8.4).
- *Stitched siblings:* cross-boundary cliques are materialized using a shared scale $\tau^* = \sqrt{\tau_C^* \tau_{C'}^*}$. Use this only when adjacent siblings' selected scales differ by less than the grid ratio $r = 1/\sqrt{2}$ (SI S2.1); otherwise the disjoint variant is preferred for numerical stability.

Cost of lifting every region. Lifting every region multiplies Stage 2 effort by the number of recursion-level regions (bounded above by leaves plus internal nodes of the recursion tree). In practice this is well-controlled: parent complexes are coarser than the union of their descendants, intermediate cluster sizes are typically a small fraction of the dataset (each AP step partitions the parent into multiple children), and every complex warm-starts from its parent's settled state (positions, links, attenuated moments; SI S4.4). Aggregate cost is dominated by leaf-level work plus a modest constant factor for the intermediate-level lifts.

4.2 Why Lift from Graphs to Simplices

The Stage 1 scaffold is useful for discovery but is not by itself an ideal domain for likelihoods or samples. Nodes and edges describe representatives and adjacency; they do not provide volume-bearing cells on which a density can be normalized. A simplicial complex supplies those cells. Simplex masses describe how probability is distributed over local patches, shared faces define adjacency for conservation and interpolation, and barycentric coordinates provide a natural local coordinate system for density evaluation and sampling.

This lift also separates concerns. The graph remains the source of evidence and topology, while the complex is the domain on which generative quantities live. Stage 2 does not discard the scaffold; it turns the scaffold into a richer geometric object.

4.3 Complex Construction

Proteus treats the Stage 2 complex as the top-dimensional flag complex of the current Hebbian graph, restricted by the local intrinsic-dimension estimate. A simplex exists exactly when its vertices form the required clique in the graph. Link or node edits therefore create or destroy all incident simplices automatically; ordinary node motion changes geometry but not simplex membership.

The initial complex is seeded by a cheap Greedy Chaining pass over scaffold neighborhoods. The pass verifies local cliques and deduplicates candidate simplices; it does not run Delaunay

Hierarchy comparison placeholder: root complex, intermediate complexes, and leaf complexes at their own scales.

Figure 3: A tree of complexes preserves coarse and fine structure simultaneously.

Torsion ladder placeholder: monitor, geometric refinement, torsion split, and local warp upgrade.

Figure 4: Torsion is a proposal signal. The evidence gate decides whether a proposed intervention is retained.

maintenance or global triangulation. This is sufficient because the Stage 1 graph has already learned local adjacency at the active scale. The SI owns the full initialization algorithm, heterogeneous-dimension handling, and cache invalidation rules.

4.4 Simplex Equilibrium and Node Equilibrium

Stage 2 aims to equilibrate simplex masses rather than only node hits. These are related but not identical: node hits measure Voronoi-like ownership around representatives, while simplex masses measure probability over local patches of the lifted complex. In regular interior regions, variance-cap equilibration drives node hit masses toward a common value, and Hebbian adjacency makes simplex stars approximately regular. Under those conditions, equilibrated simplex masses imply effective node equilibrium up to finite-sample and residual tolerance terms.

The residual term is meaningful rather than merely a nuisance. At boundaries or dimensionality junctions, a node’s simplex star can remain structurally irregular even after ordinary equilibration. Proteus treats this as a diagnostic signature: the model should not keep splitting forever to erase a real topological or dimensional transition.

4.5 Torsion Audit, Safe Edits, and Optional Warps

Piecewise-linear geometry is the default Stage 2 representation, but some curvature cannot be removed by local vertex motion alone. Proteus detects this with a torsion-style residual diagnostic on each simplex: if the residual mean field behaves like a conservative correction, the local PL patch is adequate; if the residual field develops a persistent rotational component, the patch is signaling non-PL structure or dimensional mismatch.

Torsion is only a proposal signal. Low torsion is monitored, moderate torsion proposes additional geometric refinement, and high torsion proposes either a localized split or a local invertible warp. Every proposed intervention is still tested by the same evidence gate used elsewhere. This keeps the torsion ladder from becoming a separate heuristic model-selection rule.

Proteus does not maintain an expensive per-simplex shape-quality audit. In regular interiors, variance-cap equilibration makes sliver simplices rare; residual cases localize to boundaries and dimensionality junctions, which have their own diagnostics. The one cheap guard is at split placement: if a torsion-aligned split would produce a child with edge-length ratio $\ell_{\max}/\ell_{\min} > 5$, the split falls back to a centroid star split.

Optional warps are capacity upgrades, not replacements for the complex. A warp is attached to a region or patch when curvature exceeds what the PL representation can express at the current scale and the evidence gate supports the additional capacity. At non-root levels, these warps are trained in the child PCA frame already selected by recursion, after any marginal Gaussianization in that same frame. Density and sampling still pass through the explicit complex; the warp changes the coordinate frame in which that complex is approximately Gaussian or PL-regular.

5 Evidence-Based Structure Selection

Geometry proposes; evidence accepts

The previous sections describe many mechanisms that can propose structure: variance-cap violations, weak links, simplex-star imbalances, torsion, and local warp failures. These signals are intentionally heterogeneous because they look for different kinds of strain in the model. Section 5 explains how Proteus keeps those proposal mechanisms from becoming independent model-selection rules.

The unifying principle is that every discrete edit is judged by the same local evidence comparison. The gate asks whether the candidate topology explains the routed transition evidence in its affected region better than the current topology after accounting for finite-sample complexity. Continuous quantities may guide where to look, but accepted structure must be supported by transition statistics under the same Dirichlet–multinomial criterion.

5.1 Why a Single Evidence Gate Is Needed

Proteus has many ways to notice strain: high variance can suggest a split, weak links can suggest pruning, torsion can suggest geometric remediation, and persistent curvature can suggest a warp. If each diagnostic accepted its own edits, the system would become a collection of incompatible heuristics. Instead, diagnostics only propose; a single local evidence gate decides whether the proposed topology explains routed transition statistics better than the current topology after accounting for finite-sample complexity.

This makes structural growth evidence-limited. A region can be geometrically imperfect yet still reject added complexity if the data do not support a distinct transition pattern. Conversely, a proposal from any diagnostic can be accepted when it gives a clear local likelihood improvement.

5.2 Affected Region

Discrete edits are local, so the evidence comparison is local. For a candidate topology M , the affected region $\mathcal{R}$ contains nodes whose simplex stars or neighbor sets would change, plus a small neighbor ring to stabilize router normalizers. Counts outside this region are treated as fixed boundary evidence. This keeps scoring efficient and prevents a local edit from requiring a full-model re-evaluation.

5.3 Closed-Form Dirichlet–Multinomial Score

The evidence score is a closed-form Dirichlet–multinomial marginal likelihood over outgoing transition counts. With BDeu concentration $\alpha_{0,i}$ at node i , the region score is

$$F_{\text{DM}}(\mathcal{R}; M) = - \sum_{i \in \mathcal{V}_{\text{aff}}} \log p(\{n_{i \rightarrow j}\} \mid M, \alpha_{0,i}).$$

Lower F_{DM} means higher marginal evidence. The score integrates the local categorical transition probabilities analytically, so candidates do not need iterative candidate-specific optimization of the simplex mass field. This avoids protocol-dependent biases from iteration budgets, warm starts, or partial convergence.

5.4 Acceptance Protocol and Cadence

The acceptance protocol is:

1. propose an edit from a diagnostic;
2. localize the affected region $\mathcal{R}$;

3. reroute events in $\mathcal{R}$ under the candidate topology;
4. score candidate and keep-topology alternatives with F_{DM} ;
5. accept only if the candidate improves the score by a log-Bayes-factor margin;
6. settle local continuous state before the next gate window.

In implementation, proposals are queued and scored in batches after equilibration windows. Hysteresis prevents immediate reversal of accepted edits, and per-epoch edit budgets prevent topology churn from accumulating faster than the convergence analysis allows.

5.5 Avoiding Double Counting

Both stages can change topology: Stage 1 can add or remove scaffold nodes and edges, while Stage 2 can change the induced complex through splits, prunes, merges, and warp attachments. For all such discrete edits, the acceptance evidence is the same object: routed transition counts in the affected region, re-tallied under the keep and edit candidates and scored by F_{DM} . This makes edits from different diagnostics comparable.

Simplex masses, face-pressure tallies, torsion, and warp residuals still matter, but they play different roles. They can diagnose where the model is strained, help form candidate edits, initialize or settle an accepted topology, and make the final complex usable for likelihoods and samples. They are not added as a second independent likelihood term for the same gate decision. Otherwise the same samples would influence topology twice: first through the transition router that selects structure, and again through a reconstruction conditioned on the selected structure. The rule is therefore: transition counts decide acceptance; mass and pressure fields support diagnostics, settling, and inference after acceptance.

6 Inference and Generative Interface

Memberships, likelihoods, queries, and samples

Once topology selection has terminated, Proteus has a tree of accepted complexes, each with settled geometry, simplex masses, face-pressure evidence, and optional warps. Section 6 describes how that learned object is used. The same representation supports a compact multiscale membership signal, exact PL density evaluation when needed, sampling, conditional synthesis, and user-defined region queries.

The key separation is between the fast exported membership path and the higher-fidelity mesh path. The default fuzzy membership score is a closed-form Gaussian relative-density in each region’s locally Gaussianized frame. The PL mesh remains available as the exact reference for density semantics, diagnostics, and specialized queries, but it is not required for the canonical membership vector.

6.1 Multiscale Fuzzy Memberships

For a query point $\mathbf{x}$ on recursion path $C_0 \rightarrow \dots \rightarrow C_L$, Proteus exports a scalar membership score at every level. In region C , after any attached warp T_C , the cluster is summarized by a Gaussian fit $(\boldsymbol{\mu}_C, \Sigma_C)$ in the warped frame. The canonical membership is

$$\mu_C(\mathbf{x}) = \exp\left(-\frac{1}{2}(T_C(\mathbf{x}) - \boldsymbol{\mu}_C)^\top \Sigma_C^{-1}(T_C(\mathbf{x}) - \boldsymbol{\mu}_C)\right).$$

It equals 1 at the cluster mode and decays smoothly along Mahalanobis distance. The multiscale membership trajectory is

$$\boldsymbol{\mu}(\mathbf{x}) = ((C_0, \mu_{C_0}(\mathbf{x})), \dots, (C_L, \mu_{C_L}(\mathbf{x}))).$$

Membership trajectory placeholder: root-to-leaf path with scalar Gaussian memberships at each level.

Figure 5: Multiscale fuzzy membership trajectory. Cluster identities provide the path; scalar scores provide continuous closeness to each region’s mode.

Cluster identities form an exact shared prefix for points that follow the same coarse path, while scalar scores usually differ at every level because they are evaluated continuously at the point itself.

For example, a point might have trajectory $((C_0, 0.91), (C_1, 0.64), (C_2, 0.22))$: the labels say it lies on the root-to-leaf path $C_0 \rightarrow C_1 \rightarrow C_2$, while the scalar scores say it is close to the coarse mode, moderately close to the intermediate mode, and peripheral in the leaf region.

6.2 PL Mesh as Reference and Diagnostic

The Gaussian path is the fast canonical membership interface. The PL mesh remains the exact reference for density evaluation, diagnostics, and online refinement. Disagreement between $\mu_C(\mathbf{x})$ and the PL density-derived membership $\mu_C^{\text{PL}}(\mathbf{x})$ signals incomplete Gaussianization or a region where the cached Gaussian summary should be updated. This keeps the exported membership cheap while preserving a higher-fidelity path when downstream tasks need exact mesh semantics.

6.3 Online Tallies vs. Post-Selection Reconstruction

During training, samples assigned to a simplex increment empirical face-pressure tallies $\{\hat{p}_f\}$ according to their residual orientation relative to the simplex facets. These tallies are online evidence about local flux geometry. After topology selection has terminated and the complex has settled, a separate conservative reconstruction solves for $\{p_f\}$ that both tracks the empirical tallies and satisfies discrete conservation as well as possible. The reconstructed field is an inference-time interpolant, not another structure-selection likelihood.

6.4 Density Evaluation

At a chosen level C , density evaluation locates the containing simplex S^* or projects to the nearest simplex for off-manifold queries. The local log density combines the settled simplex mass, simplex volume, and the reconstructed PL pressure profile:

$$\log p^{(C)}(\mathbf{x}) = \log m_{S^*}^{(C)} - \log |S^*|_d + \log \tilde{\rho}_{S^*}^{(C)}(\mathbf{x}) - \log \bar{w}_{S^*}^{(C)}.$$

Here $\tilde{\rho}_{S^*}^{(C)}(\mathbf{x})$ is the barycentric vertex-weight profile induced by face pressures, and $\bar{w}_{S^*}^{(C)}$ is the corresponding simplex normalizer. The SI defines the vertex-pressure aggregation, normalization, boundary conventions, and numerical volume floor in [S6.4](#), which contains the [canonical simplex-local density formula](#).

6.5 Sampling and Conditional Synthesis

Sampling at level C draws a simplex according to its settled mass, samples barycentric coordinates within that simplex, and maps the resulting point back through any attached warp. Conditional synthesis filters or reweights the candidate simplex set before sampling. Constraints can be geometric, semantic, label-derived, density-band-based, or defined by membership trajectories across multiple levels.

Algorithm 2 Inference interface (paper view)

```
1: procedure QUERYDENSITY( $\mathbf{x}, C$ )
2:   Locate simplex  $S^* = S^{(C)}(\mathbf{x})$  or nearest projected simplex
3:   Evaluate simplex-local density using  $m_{S^*}^{(C)}$ ,  $|S^*|_d$ , and the PL pressure profile
4:   return  $\log p^{(C)}(\mathbf{x})$ 
5: end procedure
6: procedure SAMPLE( $C$ )
7:   Draw simplex  $S$  proportional to  $m_S^{(C)}$ 
8:   Draw barycentric coordinates in  $S$ 
9:   Map through any inverse warp attached to  $C$ 
10:  return sample  $\mathbf{x}$ 
11: end procedure
12: procedure CONDITIONALSAMPLE( $C, \mathcal{C}$ )
13:  Restrict or reweight candidate simplices by constraint  $\mathcal{C}$ 
14:  Sample as in SAMPLE( $C$ )
15: end procedure
```

6.6 Custom Region Queries

Custom region queries are secondary to the mainline membership vector but useful for analysis. A user-defined region may be a set of simplices, a label class, a topological component, or a density band. Proteus evaluates membership in such a region by aggregating posterior simplex mass or density over the restricted set. These custom fuzzy sets compose with the multiscale trajectory, but they are not the default definition of fuzzy manifold membership.

6.7 Algorithmic Summary

7 Theoretical and Operational Status

What is proved, what is conditional, and what is operational

The preceding sections define the algorithmic objects and how they interact. This section summarizes the status of the supporting arguments without reproducing the SI proofs. The goal is to make clear which claims are mathematical reductions or finite-sample statements, which are conditional convergence results, and which are operational safeguards intended to make the first implementation robust.

The guarantees are deliberately tiered. Some statements are clean in fixed or finite-candidate settings; others depend on standard stochastic-approximation regularity or on monitorable implementation rules such as edit budgets and hysteresis. Where a result is operational rather than fully analytic, the paper states the diagnostic that is actually measured and the failure mode it is meant to control.

7.1 Scale Discovery and SVD Limit

Stage 1's moment tracking is closely related to Gaussian-windowed local PCA/SVD in the limit of fixed topology, small learning rates, high ANN recall, and sufficiently smooth sampling. The reduction explains why Oja directions and local covariance traces recover intrinsic variance directions. Proteus differs from direct SVD because it couples those local estimates to adaptive node placement, finite-data gates, and recursive scale selection.

7.2 Evidence Consistency

For any fixed affected region and finite candidate set, the Dirichlet–multinomial evidence gate is consistent under router identifiability, mixing transition events, clipped positive probabilities, and a positive expected likelihood margin. As affected-region evidence grows, the likelihood margin dominates the fixed Bayes-factor threshold and the Occam term.

7.3 Simplex–Node Correspondence

The SI proves a finite-sample correspondence from simplex equilibrium to effective node equilibrium in [S9.3](#), with the [three-term bound \(structural, equilibrium, sampling\)](#). In regular interiors, variance-cap equilibration and Hebbian adjacency drive the structural residual toward zero. Persistent residuals localize at boundaries and dimensionality junctions, where they become operational diagnostics rather than failures to be eliminated.

7.4 Dynamic Stability

The clean dynamic result is a fixed-topology, frozen-count theorem: under bounded innovations, active-node recurrence, strict cap margin, and controlled rank bias, the constant-gain dynamics enter a small invariant neighborhood of a local minimum of the vertex potential. The full streaming/editing result is conditional on standard stochastic-approximation regularity plus the finite edit-budget and settle/hysteresis rules. The claim is neighborhood stability, not Robbins–Monro point convergence.

7.5 Operational Diagnostics

Several guarantees are operational rather than fully analytic. The junction detector freezes stars whose incidence counts, intrinsic dimensions, directional asymmetries, split rejection rates, and torsion persistently indicate a dimensionality junction. The expressivity ledger tracks mass CV, junction residual, torsion percentile, held-out likelihood improvement, and sampling scale. A level stops when these diagnostics are below threshold for consecutive gate windows or when residuals are classified as junction/boundary structure.

8 Evaluation

Model quality and ablations

Proteus should be evaluated as an explicit geometric generative representation, not only as a black-box predictor. The central questions are whether the learned hierarchy recovers meaningful scale structure, whether the complexes preserve topology and geometry, whether the reconstructed fields support likelihoods and samples, and whether the membership trajectories are stable enough to be useful downstream.

For that reason, the evaluation is organized around the objects the method claims to learn. Synthetic benchmarks provide ground truth for topology, curvature, scale, and dimensionality-junction behavior; real benchmarks test whether the same diagnostics remain useful when no exact manifold is known. Ablations should remove one architectural mechanism at a time so failures can be traced to scale discovery, evidence gating, torsion diagnostics, dual-flow reconstruction, optional warps, or the decision to keep a full tree of complexes.

8.1 Experimental Setup

Evaluation should use synthetic data with known topology and geometry as well as real benchmarks without ground-truth manifolds. Synthetic cases should include variable-density manifolds,

nested or linked supports, dimensionality junctions, and curved structures requiring optional warps. Unless otherwise stated, experiments use train/test splits, multiple random seeds, and logged preprocessing decisions so scale discovery and evidence gating can be audited.

8.2 Metrics

Metrics should correspond to owned objects in the architecture:

- topological faithfulness of the learned complexes;
- reconstruction or projection error to the learned support;
- held-out log-likelihood under the reconstructed field;
- sample quality and distributional similarity;
- membership trajectory stability under resampling.

8.3 Baselines

Baselines should cover the common alternatives Proteus aims to unify: embedding-plus-density pipelines, explicit graph/topology methods without an integrated generative interface, and end-to-end deep generative models. The comparison should emphasize what object is learned and what queries it supports, not only scalar performance.

8.4 Ablations

Core ablations are: fixed scale instead of the scale controller; heuristic edit acceptance instead of the evidence gate; no torsion audit; no optional warps; uniform simplex density instead of dual flow; and leaf-only complexes instead of a full tree of complexes. Each ablation should target a named architectural claim.

9 Limitations and Scaling

9.1 Limitations

Proteus trades simplicity for explicitness. It has more moving parts than an embedding or a monolithic generative model, and first implementations must carefully maintain ownership boundaries between counts, masses, pressures, and warps. The full streaming/editing convergence story is conditional; the cleanest rigorous result is the fixed-topology tier. Junction handling is operationally specified but not yet reduced to a universal closed-form lower bound. Optional warps add training cost and should remain evidence-gated rather than automatic.

The tunable thresholds and operating defaults are consolidated in SI S14.3 so implementations can distinguish theory-derived constants from empirical first-pass settings.

9.2 Complexity by Owned Object

Complexity should be reported by owned object. Stage 1 scales with samples, ANN queries, and scaffold size. Complex construction scales with local graph neighborhoods and simplex count. Evidence scoring scales with the affected region of each proposal. Torsion and edge-ratio checks are simplex-local. Dual flow scales with the face/dual graph. Inference scales with tree descent plus one local density or sampling operation at the chosen level.

9.3 Typical Operating Envelope

The intended operating regime is data large enough to support local evidence tests, with intrinsic dimension low enough that local complexes are tractable. Very high ambient dimension should use preprocessing; very sparse regions should stop rather than recurse. Practical limits are expected to be dominated by ANN maintenance, simplex count in high intrinsic dimension, and optional warp training.

10 Conclusion

Proteus is built around a simple thesis: multiscale geometric structure should be learned as an explicit object, and structural complexity should be accepted only when finite data support it. By separating scale discovery, simplicial generative refinement, evidence-gated structure selection, and inference-time probability reconstruction, Proteus aims to provide a model that is not only generative but also inspectable, queryable, and operationally controllable.

Appendix: Notation

We summarize the core symbols used throughout the paper. Full definitions live in their owning SI sections.

- $\mathbf{m}_i \in \mathbb{R}^d$, $\mathbf{s}_i \in \mathbb{R}^d$: EWMA residual mean and squared-residual estimate at node i (SI S2.3).
- $\sigma_i^2 = \text{tr}(\mathbf{s}_i - \mathbf{m}_i \odot \mathbf{m}_i)$: scalar variance proxy at node i ; $\sigma_i = \sqrt{\sigma_i^2}$.
- $\rho_i = \|\mathbf{m}_i\|_2 / (\sigma_i + \varepsilon)$: incoherence ratio.
- $\mathbf{u}_i$: dominant residual direction (Oja update).
- $\mathbf{m}_i^\pm$, $\mathbf{s}_i^\pm$, $h_i^\pm$: partition-aligned shadow moments along $\mathbf{u}_i$ (SI S2.3.2).
- h_i , $n_{i \rightarrow j}$: node hit mass and directed transition counts.
- k : rank-weighted neighborhood size (Stage 1: $k = 8$; Stage 2: $k \approx d$).
- $s_{\text{control}} \in [0, 1)$, D_{subspace} , $d_{\text{final},i}$: scale-control variable, working dimension, smoothed local intrinsic dimension (diagnostic).
- $\tau = -D_{\text{subspace}} \log(1 - s_{\text{control}})$: variance cap, uniform within a scaffold run; $\tau_{\text{local},i} = \tau$ for all i .
- $\Phi_C(\tau)$, $R_i(\tau)$, $V_C(\tau)$: cluster-local scale response, per-node Lindeberg-normalized response, support trace (SI S2.5).
- $\Omega_S = M^\top E - E^\top M$, $\kappa_S = \|\Omega_S\|_F$, $R_S = \kappa_S / \tau^*$: simplex torsion form, scalar magnitude, dimensionless ratio.
- $m_S^{(C)}$, $\hat{p}_f^{(C)}$, $p_f^{(C)}$: simplex mass, online face-pressure tally, conservative face pressure (SI S6).
- μ_C , Σ_C : cluster MLE Gaussian fit; T_C : optional region warp.
- $\mu_C(\mathbf{x})$, $\mu_C^{\text{PL}}(\mathbf{x})$, $\mu(\mathbf{x})$: closed-form Gaussian membership, PL reference membership, multiscale trajectory (SI S7).
- $\alpha_{0,i} = 1 / (d_{\text{final},i} + 1)$: BDeu concentration (SI S2.7).
- $F_{\text{DM}}(\mathcal{R}; M)$, τ_{BF} : Dirichlet–multinomial evidence score and Bayes-factor margin (SI S3.4).

- W , T_{hyst} , A_T : gate-window length, hysteresis half-life, cumulative positive-jump budget (SI S3.6).
- J_i : vertex junction score (SI S8.4).

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